

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Barium, AAS standard solution, Specpure®, Ba 1000µg/ml**  
Cat No. : **88052**  
Molecular Formula **Ba(NO<sub>3</sub>)<sub>2</sub> in 5% HN O<sub>3</sub>**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Substances/mixtures corrosive to metal

Category 1 (H290)

### Health hazards

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Skin Corrosion/Irritation

Category 1 (H314) B

Serious Eye Damage/Eye Irritation

Category 1 (H318)

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

H290 - May be corrosive to metals

H332 - Harmful if inhaled

H314 - Causes severe skin burns and eye damage

### Precautionary Statements

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

## 2.3. Other hazards

No information available

This product does not contain any known or suspected endocrine disruptors

Toxic to terrestrial vertebrates

## **Section 3: Composition/information on ingredients**

### 3.2. Mixtures

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| Component      | CAS No     | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                        |
|----------------|------------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------|
| Water          | 7732-18-5  | 231-791-2         | 94.81    | -                                                                                                                        |
| Nitric acid    | 7697-37-2  | 231-714-2         | 5.00     | Ox. Liq. 3 (H272)<br>Met. Corr. 1 (H290)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1A (H314)<br>Eye Dam. 1 (H318)<br>(EUH071) |
| Barium nitrate | 10022-31-8 | EEC No. 233-020-5 | 0.19     | Ox. Sol. 2 (H272)<br>Acute Tox. 3 (H301)<br>Acute Tox. 4 (H332)<br>Eye Irrit. 2 (H319)                                   |

| Component   | Specific concentration limits (SCL's)                                                                                                                                                                                                                                             | M-Factor | Component notes |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|
| Nitric acid | Ox. Liq. 2 :: C>=99%<br>Ox. Liq. 3 :: 65%<=C<99%<br>Acute Tox. 1 (inhal) :: C>=70%<br>Acute Tox. 3 (inhal) ::<br>70%>C>=26.5%<br>Acute Tox. 4 (inhal) ::<br>26.5%>C>=13.25%<br>Skin Corr. 1A :: C>=20%<br>Skin Corr. 1B :: 5%<=C<20%<br>Met. Corr. 1 :: C>=2%<br>EUH071 :: C>=20% | -        | -               |

## Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

| Component   | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-------------|-----------------------|-------------------------|-----------------------------|
| Nitric acid | -                     | -                       | ATE = 2.65 mg/L (vapours)   |

ECHA (RAC) - Committee for Risk Assessment - European CHemicals Agency  
ATE - Acute Toxicity Estimate

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                       |                                                                                                                                                                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b> | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                        |
| <b>Ingestion</b>      | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                           |
| <b>Inhalation</b>     | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. |

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**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## **4.2. Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **Section 5: Firefighting measures**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Not combustible. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

#### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Barium oxides.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **Section 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### **6.2. Environmental precautions**

Should not be released into the environment.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

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## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 8B

**Switzerland - Storage of hazardous substances**

Storage class - SC 8  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component      | European Union                                             | The United Kingdom                                                    | France                                                                                              | Belgium                                                                | Spain                                                                                          |
|----------------|------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Nitric acid    | STEL: 1 ppm (15min)<br>STEL: 2.6 mg/m <sup>3</sup> (15min) | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min              | STEL / VLCT: 1 ppm.<br>indicative limit<br>STEL / VLCT: 2.6<br>mg/m <sup>3</sup> . indicative limit | STEL: 1 ppm 15<br>minuten<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minuten | STEL / VLA-EC: 1 ppm<br>(15 minutos).<br>STEL / VLA-EC: 2.6<br>mg/m <sup>3</sup> (15 minutos). |
| Barium nitrate | TWA: 0.5 mg/m <sup>3</sup> (8hr)                           | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr | TWA / VME: 0.5 mg/m <sup>3</sup><br>(8 heures). indicative<br>limit                                 |                                                                        | TWA / VLA-ED: 0.5<br>mg/m <sup>3</sup> (8 horas)                                               |

| Component      | Italy                                                                                        | Germany                                                                              | Portugal                                                                                     | The Netherlands                                                          | Finland                                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitric acid    | STEL: 1 ppm 15 minuti.<br>Short-term<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minuti. Short-term | TWA: 1 ppm (8<br>Stunden). AGW -<br>TWA: 2.6 mg/m <sup>3</sup> (8<br>Stunden). AGW - | STEL: 1 ppm 15<br>minutos<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minutos<br>TWA: 2 ppm 8 horas | STEL: 0.5 ppm 15<br>minuten<br>STEL: 1.3 mg/m <sup>3</sup> 15<br>minuten | TWA: 0.5 ppm 8<br>tunteina<br>TWA: 1.3 mg/m <sup>3</sup> 8<br>tunteina<br>STEL: 1 ppm 15<br>minuutteina<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minuutteina |
| Barium nitrate |                                                                                              | TWA: 0.5 mg/m <sup>3</sup> (8<br>Stunden). AGW -                                     | TWA: 0.5 mg/m <sup>3</sup> 8 horas                                                           |                                                                          | TWA: 0.5 mg/m <sup>3</sup> 8<br>tunteina                                                                                                                 |

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|  |  |                                                                                                       |  |  |  |
|--|--|-------------------------------------------------------------------------------------------------------|--|--|--|
|  |  | exposure factor 1<br>TWA: 0.5 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>Höhepunkt: 4 mg/m <sup>3</sup> |  |  |  |
|--|--|-------------------------------------------------------------------------------------------------------|--|--|--|

| Component      | Austria                                                                                    | Denmark                                                                  | Switzerland                                                                                                                           | Poland                                                                                  | Norway                                                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nitric acid    | MAK-KZGW: 1 ppm 15<br>Minuten<br>MAK-KZGW: 2.6 mg/m <sup>3</sup><br>15 Minuten             | STEL: 1 ppm 15<br>minutter<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minutter | STEL: 2 ppm 15<br>Minuten<br>STEL: 5 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 2 ppm 8 Stunden<br>TWA: 5 mg/m <sup>3</sup> 8<br>Stunden | STEL: 2.6 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 1.4 mg/m <sup>3</sup> 8<br>godzinach | TWA: 2 ppm 8 timer<br>TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 4 ppm 15<br>minutter. value<br>calculated<br>STEL: 10 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated |
| Barium nitrate | MAK-KZGW: 2 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 0.5 mg/m <sup>3</sup> 8<br>Stunden |                                                                          | STEL: 4 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 0.5 mg/m <sup>3</sup> 8<br>Stunden                                                    |                                                                                         | TWA: 0.5 mg/m <sup>3</sup> 8 timer                                                                                                                                          |

| Component   | Bulgaria                                     | Croatia                                                                              | Ireland                                                  | Cyprus                                     | Czech Republic                                                            |
|-------------|----------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------|
| Nitric acid | STEL : 1 ppm<br>STEL : 2.6 mg/m <sup>3</sup> | STEL-KGVI: 1 ppm 15<br>minutama.<br>STEL-KGVI: 2.6 mg/m <sup>3</sup><br>15 minutama. | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> 8<br>hodinách.<br>Ceiling: 2.5 mg/m <sup>3</sup> |

| Component   | Estonia                                                                      | Gibraltar                                                | Greece                                     | Hungary                                                                            | Iceland                                    |
|-------------|------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------|
| Nitric acid | STEL: 1 ppm 15<br>minutites.<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minutites. | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup> | STEL: 2.6 mg/m <sup>3</sup> 15<br>percekben. CK<br>STEL: 1 ppm 15<br>percekben. CK | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup> |

| Component   | Latvia                                                                                  | Lithuania                                  | Luxembourg                                                             | Malta                                                             | Romania                                                           |
|-------------|-----------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------|
| Nitric acid | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup><br>TWA: 0.78 ppm<br>TWA: 2 mg/m <sup>3</sup> | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup> | STEL: 1 ppm 15<br>Minuten<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>Minuten | STEL: 1 ppm 15 minuti<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minuti | STEL: 1 ppm 15 minute<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minute |

| Component      | Russia                                                        | Slovak Republic                | Slovenia                                                                                                                         | Sweden                                                                                                                                                                 | Turkey                                                            |
|----------------|---------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Nitric acid    | Skin notation<br>MAC: 2 mg/m <sup>3</sup>                     | Ceiling: 2.6 mg/m <sup>3</sup> | TWA: 1 ppm 8 urah<br>TWA: 2.6 mg/m <sup>3</sup> 8 urah<br>STEL: 1 ppm 15<br>minutah<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>minutah | Binding STEL: 1 ppm 15<br>minuter<br>Binding STEL: 2.6<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 0.5 ppm 8 timmar.<br>NGV<br>TLV: 1.3 mg/m <sup>3</sup> 8<br>timmar. NGV | STEL: 1 ppm 15 dakika<br>STEL: 2.6 mg/m <sup>3</sup> 15<br>dakika |
| Barium nitrate | TWA: 0.5 mg/m <sup>3</sup> 0239<br>MAC: 1.5 mg/m <sup>3</sup> |                                |                                                                                                                                  |                                                                                                                                                                        |                                                                   |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

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## Predicted No Effect Concentration (PNEC)

See values below.

| Component                             | Fresh water                        | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)           |
|---------------------------------------|------------------------------------|--------------------------------|--------------------|------------------------------------|------------------------------|
| Barium nitrate<br>10022-31-8 ( 0.19 ) | PNEC = 115µg/L<br>PNEC = 0.115mg/L | PNEC = 600mg/kg<br>sediment dw |                    | PNEC = 62.2mg/L                    | PNEC =<br>207.7mg/kg soil dw |

| Component                             | Marine water    | Marine water sediment | Marine water Intermittent | Food chain | Air |
|---------------------------------------|-----------------|-----------------------|---------------------------|------------|-----|
| Barium nitrate<br>10022-31-8 ( 0.19 ) | PNEC = 11.5µg/L |                       |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Neoprene       | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

In case of insufficient ventilation, wear suitable respiratory equipment

**Recommended Filter type:** Acid gases filter

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

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|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Liquid                   |                                   |
| Appearance                              | Clear                    |                                   |
| Odor                                    | No information available |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flammability (liquid)                   | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | No data available        |                                   |
| Water Solubility                        | Miscible                 |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Nitric acid                             | -2.3                     |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)  |                                   |

## 9.2. Other information

Molecular Formula Ba(NO<sub>3</sub>)<sub>2</sub> in 5% HN O<sub>3</sub>

## Section 10: Stability and reactivity

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

Strong bases.

### 10.6. Hazardous decomposition products

Nitrogen oxides (NO<sub>x</sub>). Barium oxides.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

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## Product Information

### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Category 4

## Toxicology data for the components

| Component      | LD50 Oral           | LD50 Dermal | LC50 Inhalation                 |
|----------------|---------------------|-------------|---------------------------------|
| Water          | -                   | -           | -                               |
| Nitric acid    | -                   | -           | LC50 = 2500 ppm. (Rat) 1h       |
| Barium nitrate | 50-300 mg/kg (Rat ) | -           | LC50 > 1.1 mg/L ( Rat ) 243 min |

| Component   | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-------------|-----------------------|-------------------------|-----------------------------|
| Nitric acid | -                     | -                       | ATE = 2.65 mg/L (vapours)   |

ECHA (RAC) - Committee for Risk Assessment - European CHemicals Agency

ATE - Acute Toxicity Estimate

### (b) skin corrosion/irritation;

Category 1 B

### (c) serious eye damage/irritation;

Category 1

### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

### (e) germ cell mutagenicity;

No data available

### (f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

### (g) reproductive toxicity;

No data available

### (h) STOT-single exposure;

No data available

### (i) STOT-repeated exposure;

No data available

Target Organs

None known.

### (j) aspiration hazard;

No data available

### Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## 11.2. Information on other hazards

### Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

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## Section 12: Ecological information

### 12.1. Toxicity

Ecotoxicity effects

### 12.2. Persistence and degradability

**Persistence**

Miscible with water, Persistence is unlikely, based on information available.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Nitric acid | -2.3    | No data available             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

### 12.5. Results of PBT and vPvB assessment

No data available for assessment.

### 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## Section 14: Transport information

IMDG/IMO

ALFAA88052

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**14.1. UN number** UN2031  
**14.2. UN proper shipping name** NITRIC ACID SOLUTION  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** II

## ADR

**14.1. UN number** UN2031  
**14.2. UN proper shipping name** NITRIC ACID SOLUTION  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** II

## IATA

**14.1. UN number** UN2031  
**14.2. UN proper shipping name** NITRIC ACID SOLUTION  
**14.3. Transport hazard class(es)** 8  
**14.4. Packing group** II

**14.5. Environmental hazards** No hazards identified  
**14.6. Special precautions for user** No special precautions required.  
**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component      | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|----------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Water          | 7732-18-5  | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |
| Nitric acid    | 7697-37-2  | 231-714-2 | -      | -   | X     | X    | KE-25911 | X    | X    |
| Barium nitrate | 10022-31-8 | 233-020-5 | -      | -   | X     | X    | KE-02070 | X    | X    |

| Component      | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|----------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Water          | 7732-18-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Nitric acid    | 7697-37-2  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Barium nitrate | 10022-31-8 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component   | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water       | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Nitric acid | 7697-37-2 | -                                                                   | Use restricted. See entry                                                     | -                                                                                                     |

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|                |            |   |                                                                     |   |
|----------------|------------|---|---------------------------------------------------------------------|---|
|                |            |   | 75.<br>(see link for restriction details)                           |   |
| Barium nitrate | 10022-31-8 | - | Use restricted. See entry 75.<br>(see link for restriction details) | - |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component      | CAS No     | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Major Accident<br>Notification | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Safety Report<br>Requirements |
|----------------|------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Water          | 7732-18-5  | Not applicable                                                                                  | Not applicable                                                                                 |
| Nitric acid    | 7697-37-2  | Not applicable                                                                                  | Not applicable                                                                                 |
| Barium nitrate | 10022-31-8 | Not applicable                                                                                  | Not applicable                                                                                 |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WKG Classification

Water endangering class = 1 (self classification)

| Component      | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|----------------|---------------------------------------|-------------------------|
| Nitric acid    | WGK1                                  |                         |
| Barium nitrate | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                         | Switzerland - Ordinance on the<br>Reduction of Risk from<br>handling of hazardous<br>substances preparation (SR<br>814.81) | Switzerland - Ordinance on<br>Incentive Taxes on Volatile<br>Organic Compounds (OVOC) | Switzerland - Ordinance of the<br>Rotterdam Convention on the<br>Prior Informed Consent<br>Procedure |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Nitric acid<br>7697-37-2 ( 5.00 ) | Prohibited and Restricted<br>Substances                                                                                    |                                                                                       |                                                                                                      |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

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## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H290 - May be corrosive to metals  
H332 - Harmful if inhaled  
H314 - Causes severe skin burns and eye damage  
H318 - Causes serious eye damage  
H272 - May intensify fire; oxidizer  
H301 - Toxic if swallowed  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
EUH071 - Corrosive to the respiratory tract

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Prepared By**

Health, Safety and Environmental Department

**Revision Date**

30-Nov-2024

**Revision Summary**

SDS sections updated.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and**

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## Preparations).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**